

A photograph of a modern university building with large glass windows and a white facade. In the foreground, there is a field of tall reeds and yellow dandelions. The sky is blue with a few clouds.

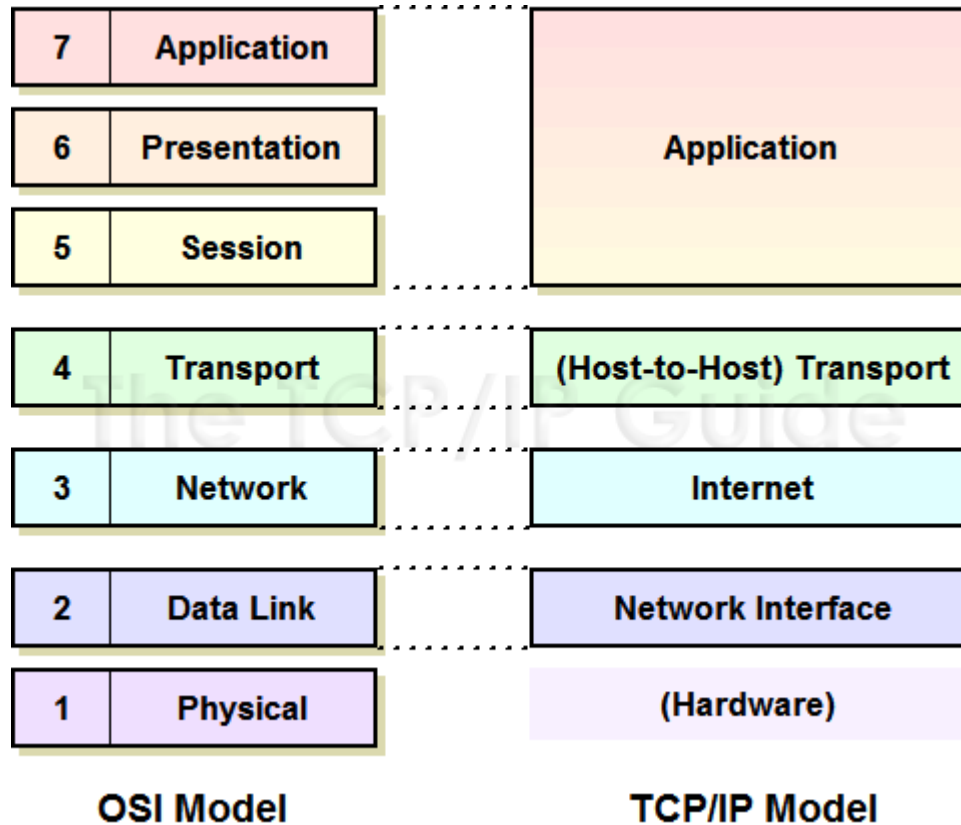
5CM512

Communication and Security Protocols

Week 5: IP Security

Class discussion

- Why do we need IP level security?



IP Security Overview

- IPsec



- “Security in the Internet Architecture”
 - Identifies key areas for security mechanisms
 - Need to secure the network infrastructure from unauthorized monitoring and control of network traffic
 - Need to secure end-user-to-end-user traffic using authentication and encryption mechanisms
 - Authentication and encryption are included as necessary security features in IPv6
 - The IPsec specification now exists as a set of Internet standards

Applications of IPsec

- IPsec provides the capability to secure communications across a LAN, private and public WANs, and the Internet



Examples include:

- Secure branch office connectivity over the Internet
- Secure remote access over the Internet
- Establishing extranet and intranet connectivity with partners
- Enhancing electronic commerce security

Class discussion

- Discuss how IPSec helps on these application examples :



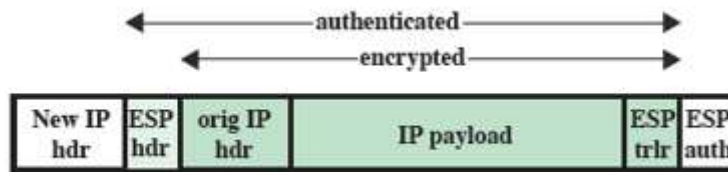
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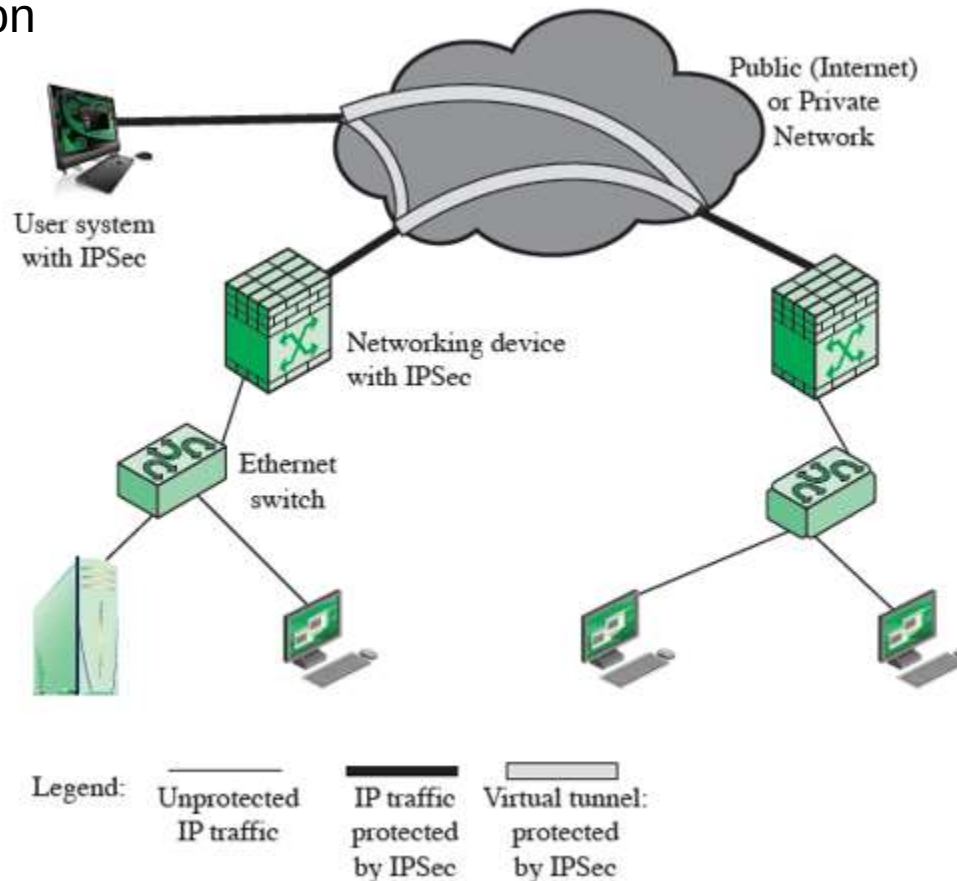
- What is the Principal Feature of IPSec?
- Principal feature of IPsec is that it can encrypt and/or authenticate all traffic at the IP level
 - Thus all distributed applications (remote logon, client/server, e-mail, file transfer, Web access) can be secured

An IPsec VPN Scenario

Tunnel-mode format



Example Configuration



Benefits of IPSec



- Implemented in a firewall or router,
 - Strong security that can be applied to all traffic crossing the perimeter
 - Traffic within a company or workgroup does not incur the overhead of security-related processing
- IPsec in a firewall
 - Resistant to bypass if all traffic from the outside must use IP
 - Only means of entrance from the Internet into the organization
- Below the transport layer (TCP, UDP)
 - Transparent to applications
 - There is no need to change software on a user or server system

Benefits of IPSec

- Transparent to end users
 - No need to
 - Train users on security mechanisms,
 - Issue keying material on a per-user basis
 - Revoke keying material when users leave the organization
- Provide security for individual users if needed
 - Useful for
 - Offsite workers
 - Setting up a secure virtual subnetwork within an organization for sensitive applications

Routing Applications

- IPsec can play a vital role in the routing architecture required for internetworking

IPsec can assure that:

A router advertisement comes from an authorized router

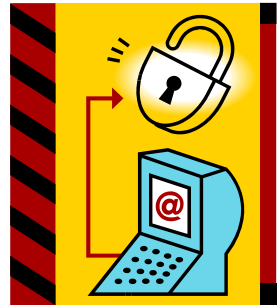
A router seeking to establish or maintain a neighbor relationship with a router in another routing domain is an authorized router

A redirect message comes from the router to which the initial IP packet was sent

A routing update is not forged

IPsec Services

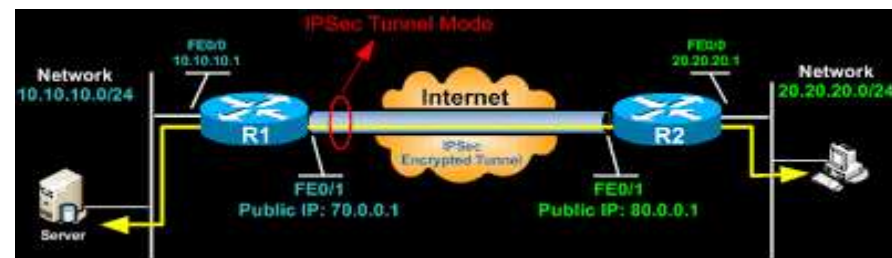
- IPsec provides security services at the IP layer by enabling a system to:
 - Select required security protocols
 - Determine the algorithm(s) to use for the service(s)
 - Put in place any cryptographic keys required to provide the requested services
- RFC 4301 lists the following services:
 - Access control
 - Connectionless integrity
 - Data origin authentication
 - Rejection of replayed packets (a form of partial sequence integrity)
 - Confidentiality (encryption)
 - Limited traffic flow confidentiality



Transport and Tunnel Modes

Transport Mode

- Provides protection primarily for upper-layer protocols
- Examples include a TCP or UDP segment or an ICMP packet
- Typically used for end-to-end communication between two hosts
- ESP in transport mode encrypts and optionally authenticates the IP payload but not the IP header
- AH in transport mode authenticates the IP payload and selected portions of the IP header



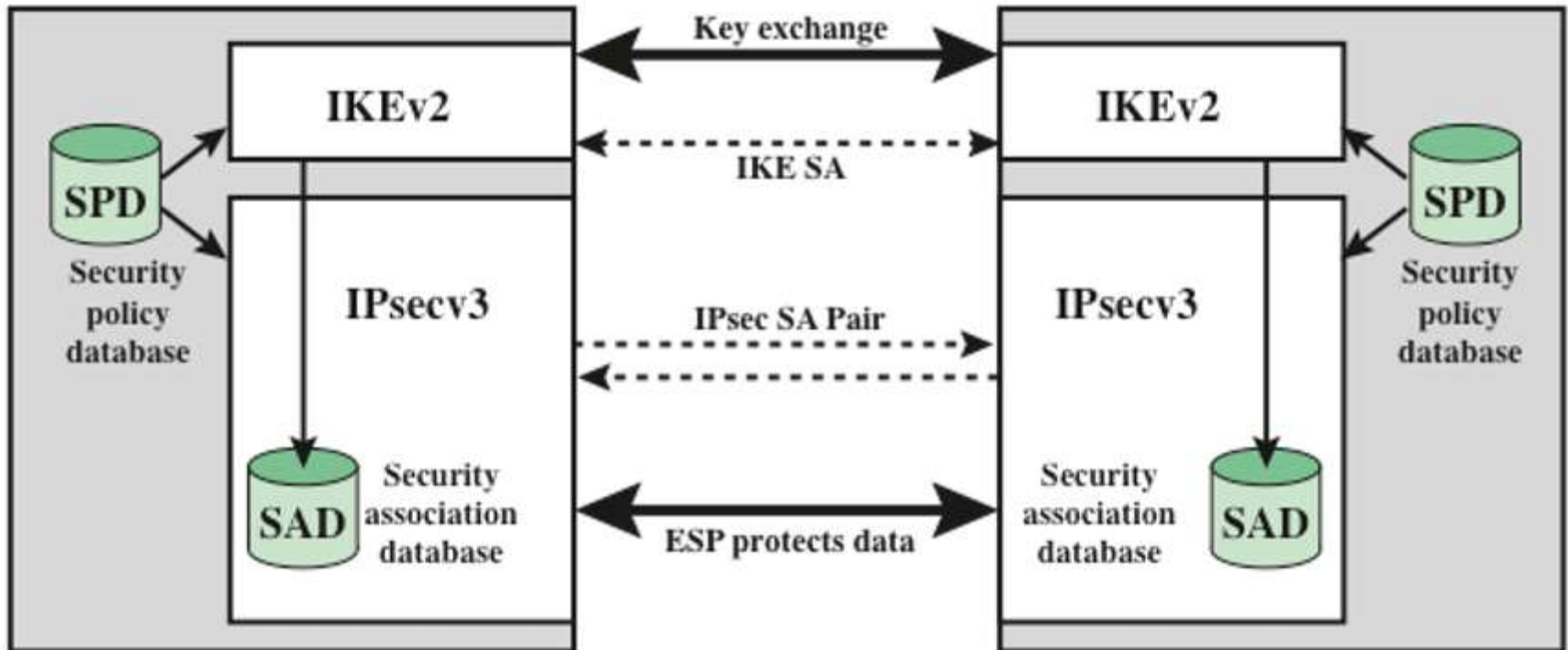
Tunnel Mode

- Provides protection to the entire IP packet
- Used when one or both ends of a security association (SA) are a security gateway
- A number of hosts on networks behind firewalls may engage in secure communications without implementing IPsec
- ESP in tunnel mode encrypts and optionally authenticates the entire inner IP packet, including the inner IP header
- AH in tunnel mode authenticates the entire inner IP packet and selected portions of the outer IP header

Tunnel Mode and Transport Mode Functionality

	Transport Mode SA	Tunnel Mode SA
AH	Authenticates IP payload and selected portions of IP header and IPv6 extension headers.	Authenticates entire inner IP packet (inner header plus IP payload) plus selected portions of outer IP header and outer IPv6 extension headers.
ESP	Encrypts IP payload and any IPv6 extension headers following the ESP header.	Encrypts entire inner IP packet.
ESP with Authentication	Encrypts IP payload and any IPv6 extension headers following the ESP header. Authenticates IP payload but not IP header.	Encrypts entire inner IP packet. Authenticates inner IP packet.

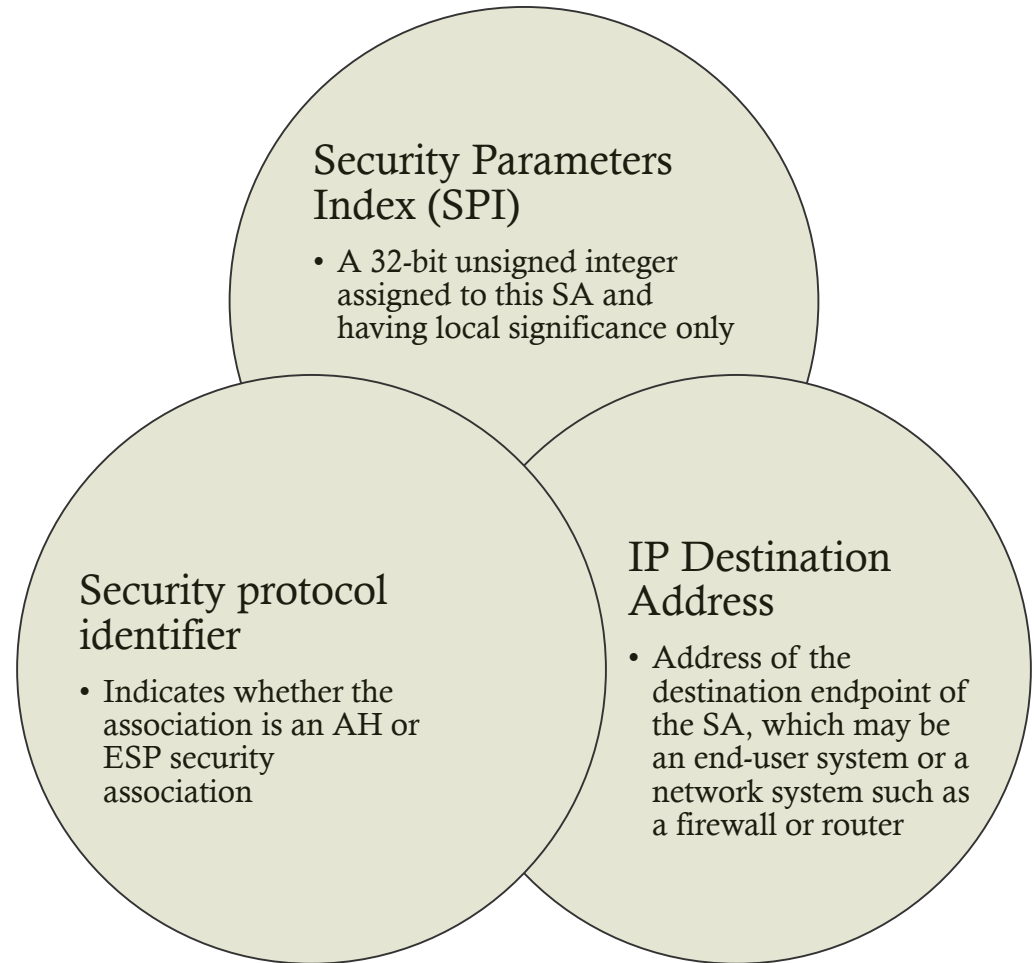
IPsec Architecture



Security Association (SA)

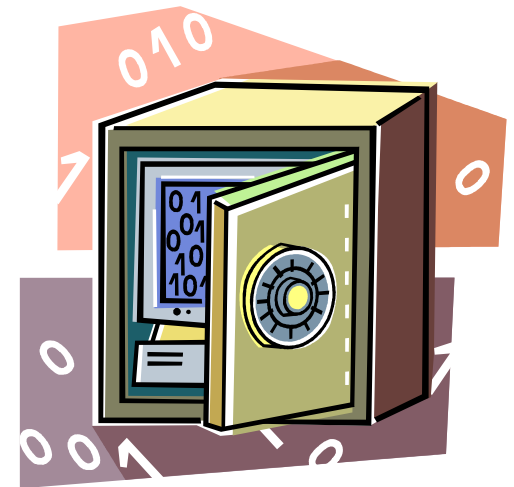
Uniquely identified by three parameters:

- A one-way logical connection between a sender and a receiver that affords security services to the traffic carried on it
- In any IP packet, the SA is uniquely identified by the Destination Address in the IPv4 or IPv6 header and the SPI in the enclosed extension header (AH or ESP)



Security Association Database (SAD)

- Defines the parameters associated with each SA
- Normally defined by the following parameters in a SAD entry:
 - Security parameter index
 - Sequence number counter
 - Sequence counter overflow
 - Anti-replay window
 - AH information
 - ESP information
 - Lifetime of this security association
 - IPsec protocol mode
 - Path MTU



Security Policy Database (SPD)

- The means by which IP traffic is related to specific SAs
 - Contains entries, each of which defines a subset of IP traffic and points to an SA for that traffic
- In more complex environments, there may be multiple entries that potentially relate to a single SA or multiple SAs associated with a single SPD entry
 - Each SPD entry is defined by a set of IP and upper-layer protocol field values called *selectors*
 - These are used to filter outgoing traffic in order to map it into a particular SA

SPD Entries

- The following selectors determine an SPD entry:

Remote IP address

This may be a single IP address, an enumerated list or range of addresses, or a wildcard (mask) address

The latter two are required to support more than one destination system sharing the same SA

Local IP address

This may be a single IP address, an enumerated list or range of addresses, or a wildcard (mask) address

The latter two are required to support more than one source system sharing the same SA

Next layer protocol

The IP protocol header includes a field that designates the protocol operating over IP

Name

A user identifier from the operating system

Not a field in the IP or upper-layer headers but is available if IPsec is running on the same operating system as the user

Local and remote ports

These may be individual TCP or UDP port values, an enumerated list of ports, or a wildcard port

Class discussion

- Discuss the following Host SPD Example

Protocol	Local IP	Port	Remote IP	Port	Action	Comment
UDP	1.2.3.101	500	*	500	BYPASS	IKE
ICMP	1.2.3.101	*	*	*	BYPASS	Error messages
*	1.2.3.101	*	1.2.3.0/24	*	PROTECT: ESP intransport-mode	Encrypt intranet traffic
TCP	1.2.3.101	*	1.2.4.10	80	PROTECT: ESP intransport-mode	Encrypt to server
TCP	1.2.3.101	*	1.2.4.10	443	BYPASS	TLS: avoid double encryption
*	1.2.3.101	*	1.2.4.0/24	*	DISCARD	Others in DMZ
*	1.2.3.101	*	*	*	BYPASS	Internet

Any QUESTIONS?

